

PLANTS AND VEGETATION AS FUELS



WHAT ARE FUELS?

Plant *fuels* are named for the role they play when they burn in a vegetation fire.

In a fire science context, fuels are *any combustible plant material*. In an ecological context, they are the totality of live and dead plant organic matter, or *phytobiomass*.

The amount of fuel present at a site that may burn in the most intense fire expectable at a given location is called *potential fuel*. For example, in a forest fire, the trunks and thicker branches of mature trees are too thick to burn.

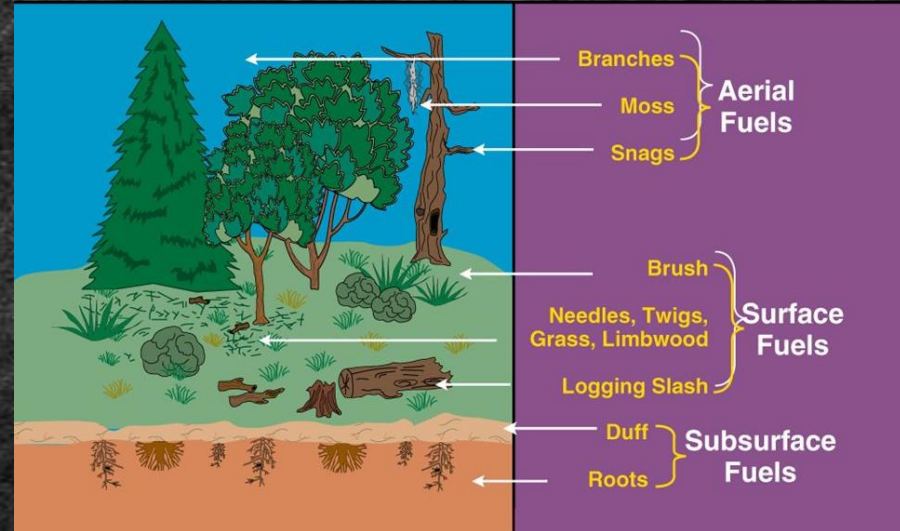
Fuels are the most important environmental factor in fire management because, unlike the weather and topography, we can modify them.

BASIC TERMINOLOGY

A **fuel complex** is the array of biomass types for a given area, an association of fuel of various **ground**, **surface** and **aerial** components. It is the coarsest scale of fuel description.

A **fuelbed**, or **fuel layer** is an array of fuels described by its loading, depth and particle sizes.

The finest scale of fuelbed description is the **fuel particle**, which defines a specific piece of fuel that is part of a fuelbed. For example, a fuel particle can be an intact or fragmented stick, grass blade, shrub leaf, or pine needle.



BASIC TERMINOLOGY

Fuel Type. An identifiable association of fuel elements of a distinctive plant species, form, size, arrangement, or other characteristics that will cause a predictable rate of fire spread or difficulty of control under specified weather conditions.

Fuel Bed. An array of fuels described by its loading, depth and particle size, commonly used to describe the fuel composition in natural settings.

Fuel Loading. The amount of fuel present at a given time and location, expressed quantitatively in terms of weight of fuel per unit area (e.g. kg.m^{-2} , or tons.ha^{-1}).

Fuel complex is an association of fuel components—usually variable, mosaic, mix of various ground, surface and aerial components.

BASIC TERMINOLOGY

Ground fuels are all organic matter below the ground line. In the figure on the previous slide, **litter** (PT: *folhada*) is considered a surface fuel, while **duff** (PT: *detritos em decomposição*) is a ground fuel.

Surface fuels are all biomass within 2 m above the ground surface.

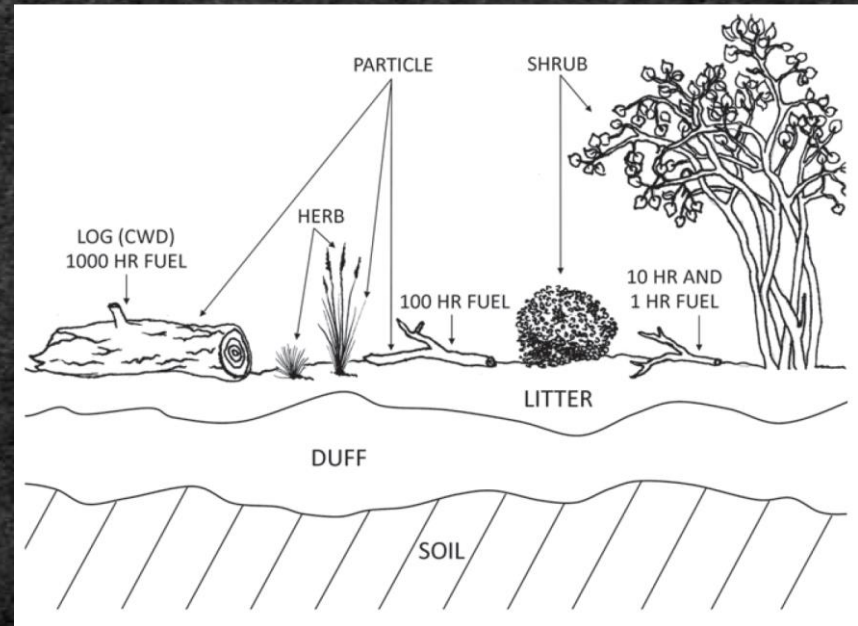
Canopy fuels are the biomass above the surface fuel layer. They are defined as all biomass (e.g., shrub, moss, lichen, vine, dead material, and tree) that is higher than 2 m above the ground surface. The term **aerial fuel** is also used to describe canopy fuel.



BASIC TERMINOLOGY

Fuel layers are composed of finer-scale elements called *fuel particles*, an intact or fragmented twig, grass blade, shrub leaf, or pine needle.

Fuel types are further subdivided into *size classes*.



DEAD FUEL MOISTURE

Fuel status: moisture

Dynamics

Phenology

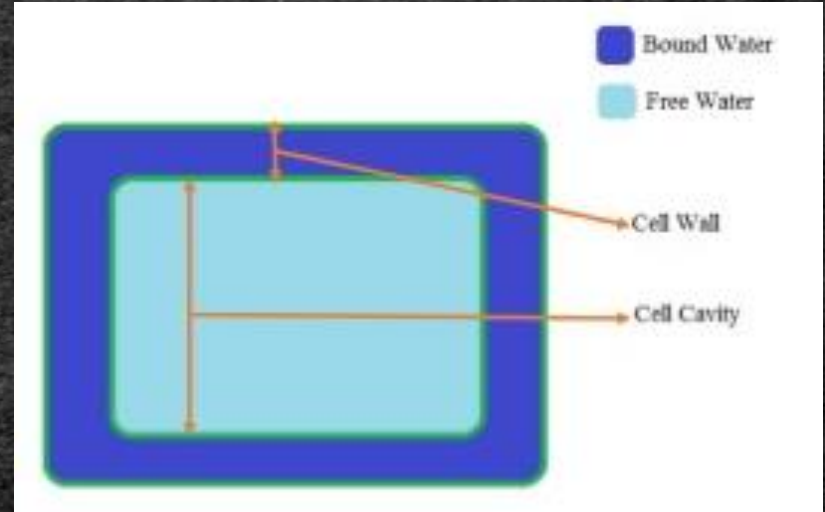
Live vs dead

DEAD FUEL MOISTURE

Water, or moisture, exists in **dead woody fuels** in two forms: **bound water** and **free water**.

Bound water is attached to cell walls by chemical bonding (H-bond). It is held within the cell walls by bonding forces between water and hollocellulose (cellulose + hemicellulose) molecules.

Free water is not chemically bonded with the cell wall and is contained in the cell cavities (*lumen*). It is comparable to water in a pipe.

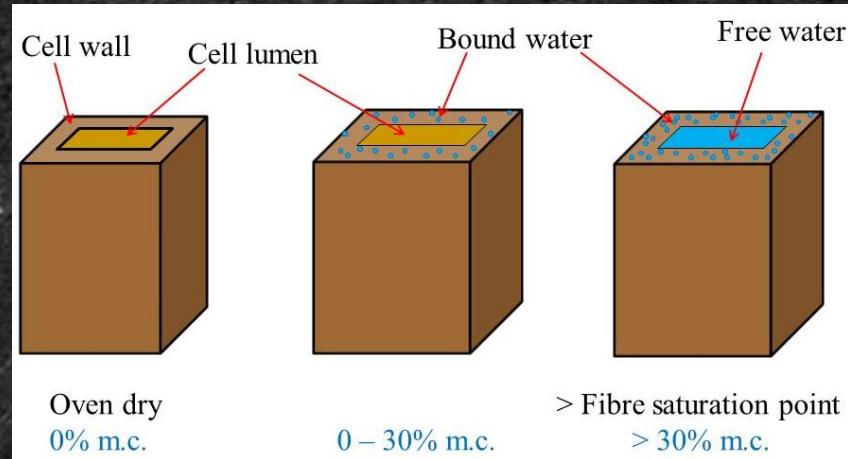


DEAD FUEL MOISTURE

The moisture content (MC) at which the cell wall is saturated with water and there is no free water in the cell lumen is called the ***fiber saturation point (FSP)***.

The FSP varies slightly between species, averaging at about ***30% moisture content***.

Below 30% MC, all the water in the dead fuel is bound water. It is only when the fuel MC exceeds 30% that free water is present.



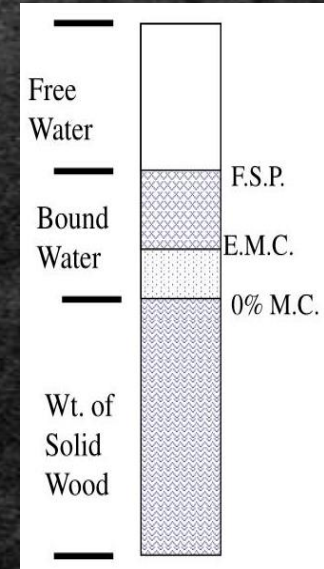
DEAD FUEL MOISTURE

As a wet, dead woody fuel dries out it first loses all free water present and only after that does it start losing the bound water.

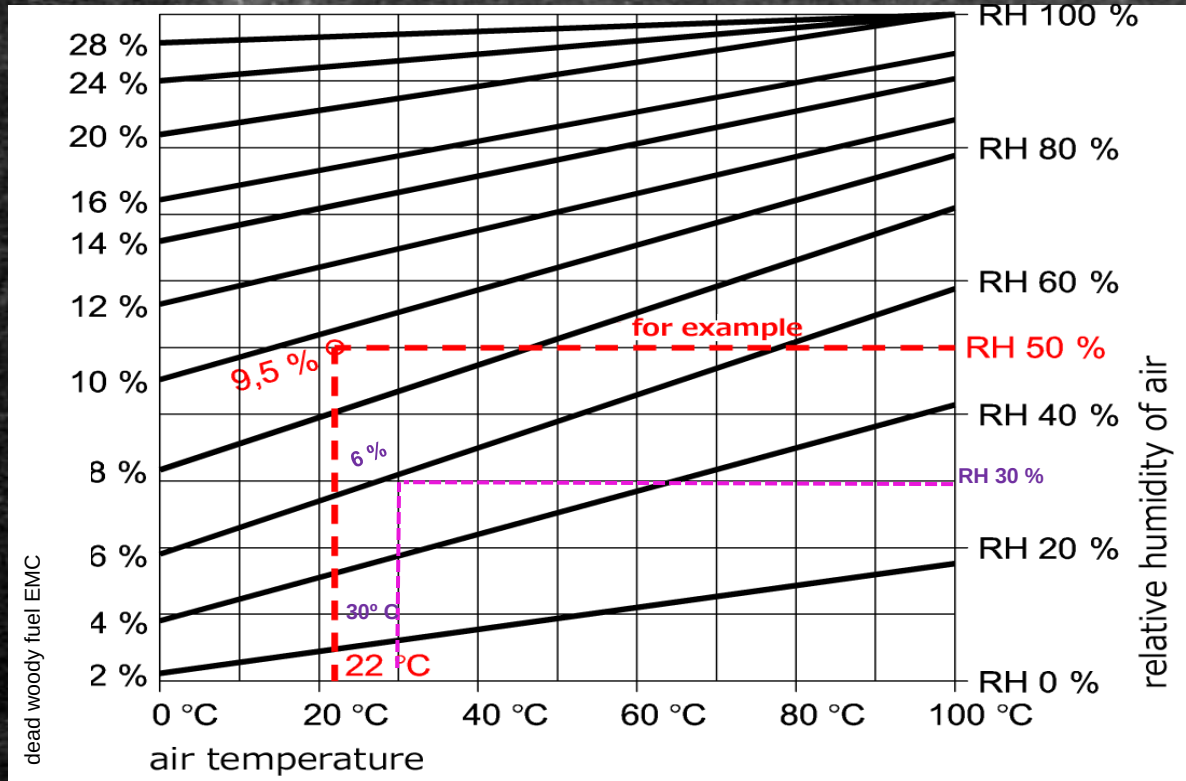
Equilibrium moisture content (EMC) is the moisture level at which the wood neither gains nor loses moisture, since it is at equilibrium with the relative humidity and temperature of the surrounding environment.

Consider a dead fuel particle with 35% MC. For a relative humidity of 60%, its EMC will be 11.5% (*see next slide*). Therefore, as it dries out, first it will lose 5% free water (MC – FSP), and then loses 18.5% bound water, for a total loss of 23.5% ($35\% - 11.5\% = \text{MC} - \text{EMC}$).

Since it takes less energy to remove free water from the wood, the initial stage of drying will be fast, but drying slows down below the FSP due to the higher energy required to break the H-bonds.



DEAD FUEL MOISTURE



In a hot (30°C) and dry (30% RH) day, the dead fuel EMC will be at a very low value of 6%.

Such a dry fuel requires little energy to ignite, generating high fire danger.

DEAD FUEL MOISTURE EXCHANGE TIMELAG

Dead fuel MC varies constantly. The fuel may gain moisture from liquid water or water vapour, while drying only involves evaporation from the fuel particle to the environment.

Dead fuel MC depends on weather variables:

Rainfall

Wind

Air temperature

Air relative humidity

Vertically-arranged fuels absorb water mostly while their surface is wet. Therefore, their MC content depends more on the duration than on the total amount of rainfall, most of which slides down the fuel to the ground.

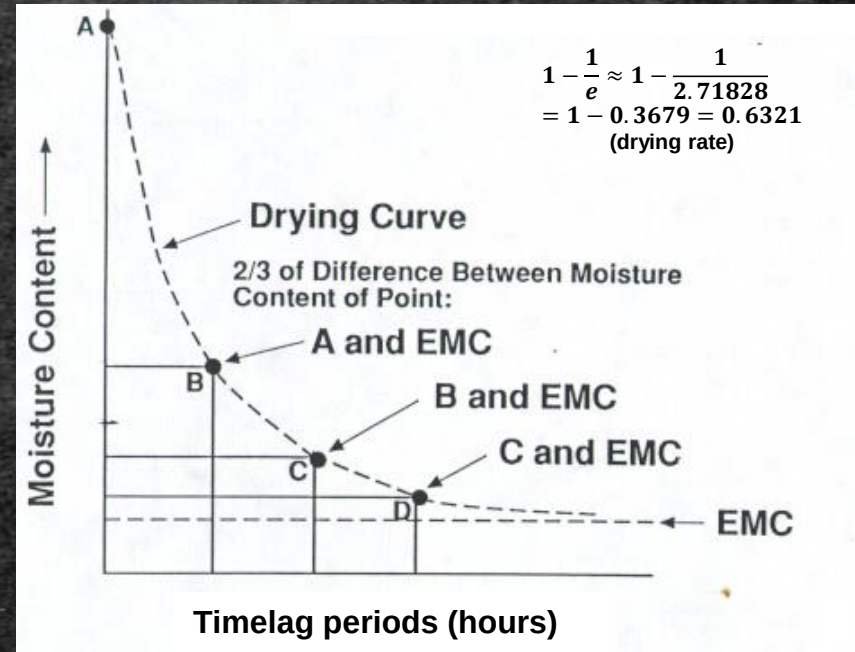
In horizontally-arranged fuels MC depends on the amount of precipitation, because the soil retains water that is continues to transfer to the fuel after the rain stops.

DEAD FUEL MOISTURE EXCHANGE TIMELAG

Dead fuels are classified as a function of the time they take to respond to variations in relative humidity of the air (RH).

Fuel drying timelag (TL, in hours) is the time required for a dead fuel to lose 63% ($1 - \frac{1}{e}$), or approximately 2/3 of the difference between its initial MC and the EMC, under constant reference experimental values of $T_{\text{air}} = 27^\circ\text{C}$ and $RH_{\text{air}} = 20\%$.

The **faster initial drying rate** corresponds to the easier removal of **free water**, followed by **slower rates** associated with the removal of **bound water**.



DEAD FUEL MOISTURE EXCHANGE TIMELAG

Período TR 1
(A-B)

$$28\% - 5.5\% = 22.5\%$$

$$22.5\% \times 63\% = 14.2\%$$

$$28\% - 14.2\% = 13.8\%$$

A humidade inicial de um combustível morto (ex. ramo de urze) é 28%. Este ramo de urze existe num ambiente onde o teor de equilíbrio da humidade é 5.5%. A diferença é de 22.5%. Após um período de tempo de resposta (independentemente da dimensão do combustível) a diferença de 22.5% é multiplicada pela constante 63% e vem igual a 14.2%. O teor de humidade alcançado pelo ramo é de $28\% - 14.2\% = 13.8\%$.

Período TR 2
(B-C)

$$13.8\% - 5.5\% = 8.3\%$$

$$8.3\% \times 63\% = 5.2\%$$

$$13.8\% - 5.2\% = 8.6\%$$

Para determinar o teor de humidade após dois períodos: 13.8% menos o teor de equilíbrio da humidade de 5.5% = 8.3%. Multiplicando 8.3 pela constante 63% = 5.2%. Subtraindo 5.2% dos 13.8% iniciais dá um teor de humidade do combustível de 8.6%.

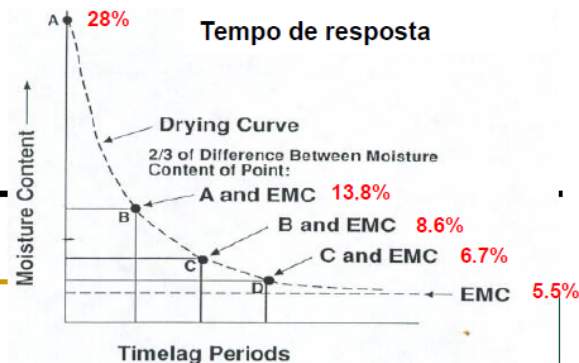
Período TR 3
(C-D)

$$8.6\% - 5.5\% = 3.1\%$$

$$3.1\% \times 63\% = 1.9\%$$

$$8.6\% - 1.9\% = 6.7\%$$

Para determinar o teor de humidade do ramo de urze após três períodos, a mesma sequência de cálculos resulta num teor de humidade de 6.7%.



DEAD FUEL MOISTURE EXCHANGE TIMELAG

Timelag depends primarily on fuel particle S/V ratio (σ), which can be estimated from particle diameter for dead woody fuels.

Fine fuels have TL periods of a few hours (i.e., they dry out and moisten up quickly in response to changes in RH_{air}), while very coarse fuels may have TL longer than one month (i.e., they take a very long time to respond to changes in RH_{air}).

The table below also shows TL for subsurface fuels, which is a function of depth.

Relationship between dead fuel particle diameter, fuel depth, and timelag.

TL (hours)	Diameter (cm)	Depth (cm)	Examples
1	< 0.6	superficial	Litter, needles, fine twigs
10	0.6 – 2.5	< 1.5	Thicker twigs, branches, bark
100	2.5 – 7.5	1.5 - 10	Thick branches
1000	> 7.5	> 10	Thick branches, tree trunks

DEAD FUEL MOISTURE EXCHANGE TIMELAG

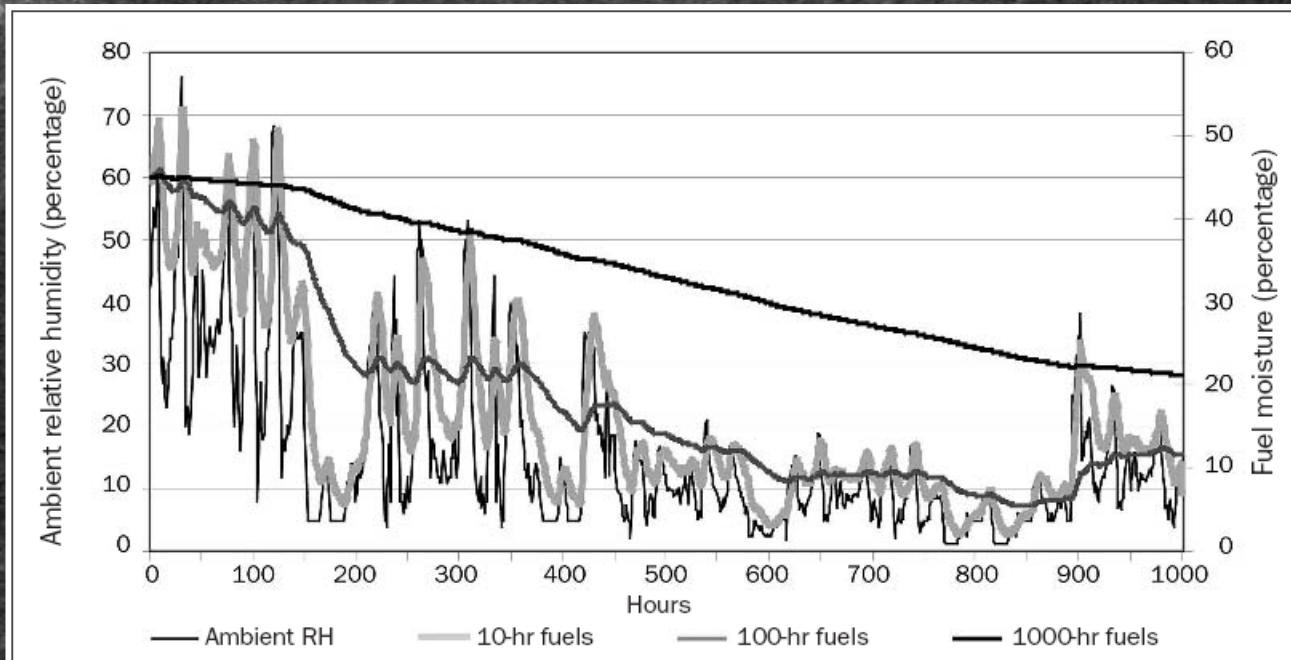


Figure 4. A theoretical example illustrating differences in fuel-moisture time lags for small (10-hour), intermediate (100-hour), and large (1000-hour) fuels. Small fuels dry out rapidly and respond more quickly to short-term variability in ambient relative humidity, while large fuels exhibit a more lagged response, requiring much longer dry periods to reach similar dryness.

SURFACE FUEL PROPERTIES

PARTICLE PROPERTIES

Particle diameter (d)

It is a critical fuel parameter for fire behavior modeling. Fuel loading (W) is stratified into particle diameter classes, which play distinct roles in the combustion and fire spread processes.

For the sake of simplicity, it is assumed that fuel particles have simple geometric shapes, like cylinders for branches and needles, and thin plates for broad leaves.

Particle diameter is divided into four size classes: < 0.6 cm, $0.6 - 2.5$ cm, $2.5 - 7.5$ cm, and > 7.5 cm.

SURFACE FUEL PROPERTIES

PARTICLE PROPERTIES

Surface-to-volume ratio (σ)

It is defined as the area of a particle surface (m^{-2}) divided by the volume of that particle (m^{-3}), and thus has units of m^{-1} . It is often indirectly estimated from particle diameter (d), using the equations:

$\sigma = \frac{4}{d}$, for approximately cylindrical objects, and

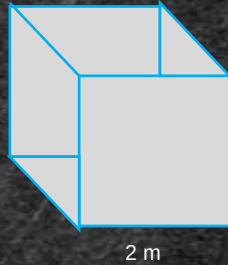
$\sigma = \frac{2}{t}$, for objects approximated as thin plates, where t is the plate thickness.

Thick particles, such as logs, *have low σ* , while *thin particles*, like grass blades or pine needles *have high σ* .

SURFACE FUEL PROPERTIES



$$\frac{S}{V} = \frac{6 * 1m^2}{1^3m^3} = 6m^{-1}$$



$$\frac{S}{V} = \frac{6 * 4m^2}{2^3m^3} = \frac{24m^2}{8m^3} = 3m^{-1}$$

These two objects have the same exact shape, but the *increase in size reduces the surface-to-volume ratio*, σ . Therefore, the larger object has a smaller surface area per unit volume through which to exchange mass (e.g. water) and energy (e.g. radiant heat) with its environment. Thus, it will take longer to moisten up / dry out, and to heat up / cool down than the smaller object.

Evidently, the same principle applies to fuel particles of different shapes, such as twigs and leaves.

SURFACE FUEL PROPERTIES

PARTICLE PROPERTIES

Surface-to-volume ratio (σ)

BOX 3.1. THE SURFACE AREA TO VOLUME RATIO

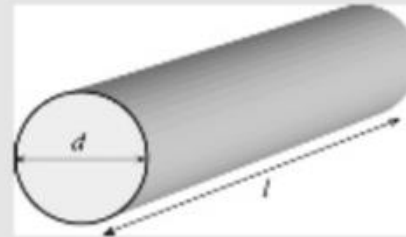
This ratio is calculated by dividing the surface area of a fuel particle by its volume:

$$SV = \frac{\pi dl}{\pi(d/2)^2 l}$$

If you ignore the ends, the equation is simplified to dividing 4 by the diameter of the particle:

$$SV = \frac{4}{d}$$

The ratio is increased if you split the fuels into smaller parts. For example, take a log with a diameter of 6:



if $d = 6$,
 $SV = 0.67$.

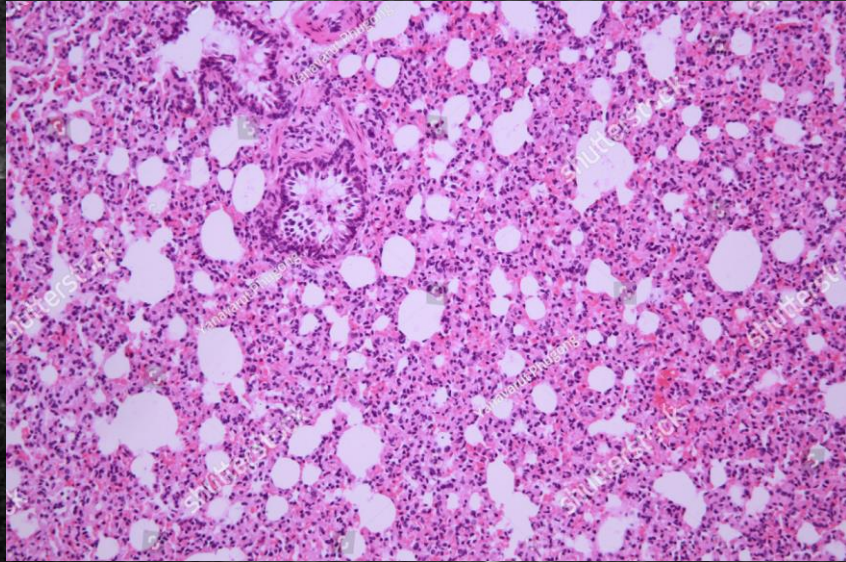
When split into 7 pieces, the ratio increases from 0.67 to 14:



if $d = 2$,
 $SV = 2$
and
 $7 \times 2 = 14$.

SURFACE FUEL PROPERTIES

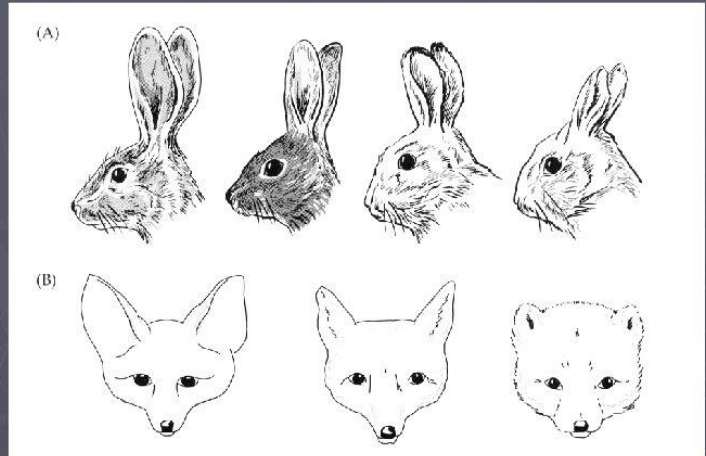
Surface-to-volume ratios in two very different biological contexts:



The human lung evolved for gaseous exchange between body and environment. Natural selection designed it to pack a lot of surface area in a small volume: $S = 100 \text{ m}^2 / V = 0.006 \text{ m}^3 \Rightarrow \sigma = 16667 \text{ m}^2/\text{m}^3$.

In hot regions, rabbits and foxes have large ears, with high σ to radiate excess heat and help cool the body. In cold regions, the animals have small ears, with low σ , to reduce heat loss.

Allen's Rule showing ear length in rabbits and foxes where the organism on the left occurs in a hot desert habitat and the one on the right occurs in a cold tundra environment



SURFACE FUEL PROPERTIES

PARTICLE PROPERTIES

Particle density (ρ_p)

Particle density is the dry weight of the particle per unit volume (kg.m^{-3}).

The term *specific gravity* is also used to represent particle density; specific gravity is the density of a substance relative to the density of water at a specific temperature and pressure. One needs to multiply specific gravity by 1000 to convert to density (e.g., 0.42 specific gravity is 420 kg m^{-3}).

EXPLAIN WHY WE CARE ABOUT IT

SURFACE FUEL PROPERTIES

PARTICLE PROPERTIES

Mineral content (S_e , S_T)

Mineral content has a profound impact on fire behavior.

The *total mineral content (S_T)* is the percent of the weight per unit volume of fuel particle that is inorganic material or mineral (i.e., not composed of molecules of C, H, and O). *Effective mineral content (S_e)* is the mineral content with the proportion of silica removed (silica does not affect the combustion reactions).

Wood is usually 1 % mineral, while the bark can have ten times that amount. Ground fuels usually have the highest mineral contents, often greater than 10 %, partially explaining why ground fuels mostly burn in smoldering combustion.

SURFACE FUEL PROPERTIES

PARTICLE PROPERTIES

Heat content (h)

The heat content (h) is the *energy released by the combustion of one unit mass of the fuel* (kJ.kg^{-1}). The heat content of wildland fuel depends on its chemical composition.

While the majority of fuel is cellulose and lignin, but other chemical constituents of fuel affect its heat content. Biomass with high mineral contents will have lower heat content (minerals do not burn, and thus do not release heat. They form the ash residue from combustion).

Other chemical compounds in biomass, like oils, resins, and proteins increase heat content in foliage and other parts of the plant.

SURFACE FUEL PROPERTIES

PARTICLE PROPERTIES

Heat content (h)

Foliage tend to have higher heat contents ($20000\text{--}21000 \text{ kJ.kg}^{-1}$) than twigs and stems ($18000\text{--}20000 \text{ kJ.kg}^{-1}$).

The heat content of wood ranges from 19300 to 22500 kJ.kg^{-1} , while that of bark is substantially higher, ranging from 20200 to 25.3 MJ kg^{-1} .

In spite of this variability, most fire models use a constant value for heat content, typically between **18000 kJ.kg^{-1} and 20000 kJ.kg^{-1}** .

SURFACE FUEL PROPERTIES

LAYER PROPERTIES

Fuel loading (W)

Loading is the *dry weight mass of the fuelbed per unit area*. It is reported in *dry weight* to eliminate the effect of variable fuel moisture content, which can vary substantially over a fire season. It is most commonly *measured in kg.m⁻²*.

Fuel loading is a key fuel property because it is used extensively in fire management. The calculation of fire intensity, one of the most important fire behavior characteristics, namely for estimating fire effects, requires an estimate of fuel loading.

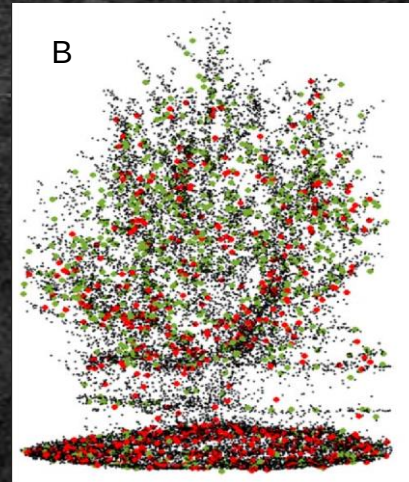
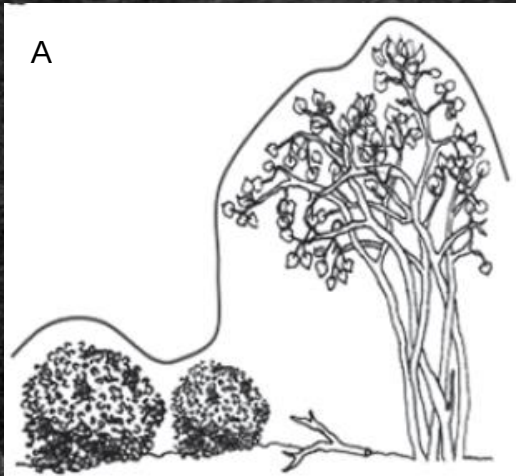
Loading is also used to estimate smoke emissions, which directly affect human health and the global climate.

SURFACE FUEL PROPERTIES

LAYER PROPERTIES

Fuel layer depth (δ)

Fuel layer depth is the thickness of the **surface** fuel layer. Traditionally, it is calculated by envisioning a virtual sheet laid over the top of the surface fuel layer and visually estimating the average height of that sheet (A).



LiDaR technology (B) allows a much more accurate estimation of fuel layer structure (B).

SURFACE FUEL PROPERTIES

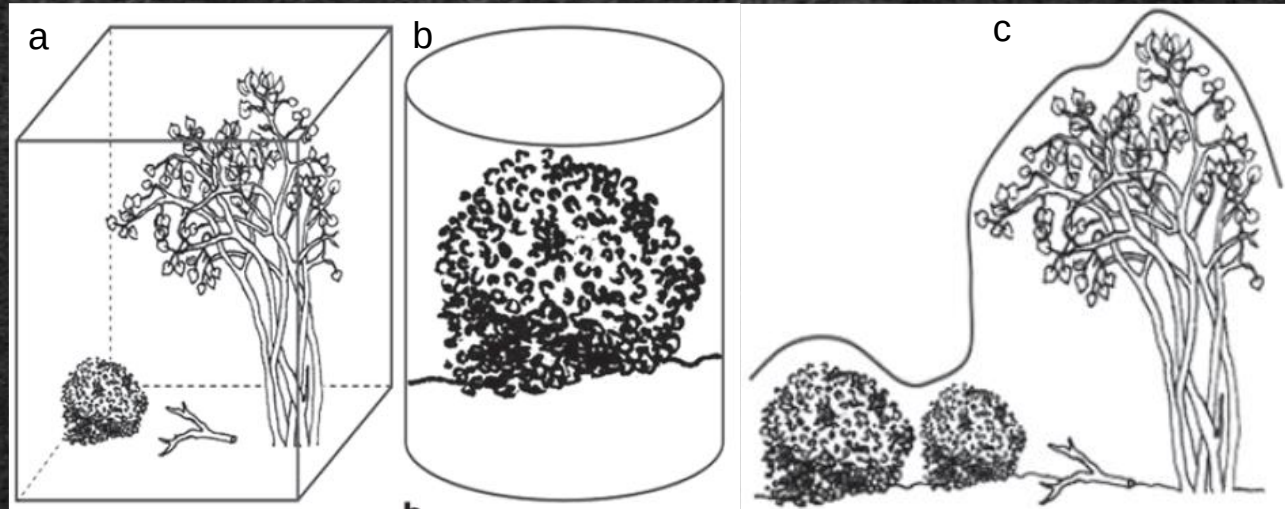
LAYER PROPERTIES

Bulk density (ρ_b)

The bulk density of a fuel component is the *mass of the fuel component material divided by the volume of space it occupies* (kg.m^{-3}).

Calculation of fuelbed volume may be performed in different ways, as shown in the figure.

- a) the volume is calculated from an estimate of depth across the fuelbed unit area;
- b) the volume is calculated for the individual plant or particle;
- c) volume is calculated as an integrated average fuelbed depth.



SURFACE FUEL PROPERTIES

LAYER PROPERTIES

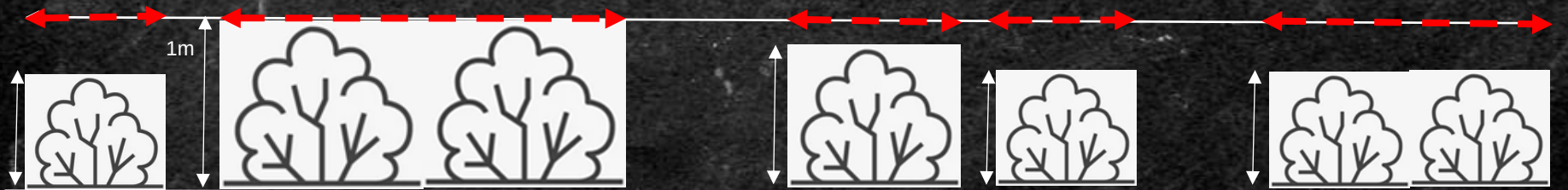
Bulk density (ρ_b)

bulk density (kg.m^{-3}) * plant cover (%) * mean height (m) = fuel loading (kg.m^{-2})

% shrub cover = 75%

mean shrub height = 0.62 m

Assuming bulk density = $2 \text{ kg.m}^{-3} \Rightarrow$ fuel loading = $0.75 * 0.62 \text{ m} * 2 \text{ kg.m}^{-3} = 0.93 \text{ kg.m}^{-2}$
 $= 9.3 \text{ t.ha}^{-1}$



SURFACE FUEL PROPERTIES

Bulk densities (kg.m^{-3})
for some Portuguese
shrub species and
genera.

Espécie	Nome Vulgar	Média 1*	Média 2*	Mediana	Desvio Padrão
<i>Phillyrea latifolia</i> L. ou <i>Rhamnus alaternus</i> L.	Adernos ¹	1,943	2,188	-	-
<i>Rosmarinus officinalis</i> L.	Alecrim ¹	1,943	2,188	-	-
<i>Pistacia lentiscus</i> L.	Aroeira ou Lentisco-verdadeiro ¹	1,943	2,188	-	-
<i>Ilex aquifolium</i> L.	Azevinho ¹	1,943	2,188	-	-
<i>Pteropartum tridentatum</i> (L.)	Carqueja ²	3,488	3,488	3,775	0,685
<i>Quercus coccifera</i> L.	Carrasco ³	1,305	1,622	1,416	0,830
<i>Quercus lusitanica</i> Lam.	Carvalhica ³	1,305	1,622	1,416	0,830
<i>Pyrus</i> spp.	Catapereiro ¹	1,943	2,188	-	-
<i>Adenocarpus</i> spp.	Codeço ⁴	1,929	3,124	2,500	3,685
<i>Cistus ladanifer</i> L.	Esteva ⁵	1,208	1,208	1,011	0,584
<i>Cytisus</i> spp., <i>Genista</i> spp. ou <i>Spartium</i> spp.	Giestas ⁴	1,929	3,124	2,500	3,685
<i>Ruscus aculeatus</i> L.	Gilbardeira ¹	1,943	2,188	-	-
<i>Phillyrea angustifolia</i> L.	Léntisco-bastardo ¹	1,943	2,188	-	-
<i>Arbutus unedo</i>	Medronheiro ⁶	1,476	1,476	1,430	0,777
<i>Lavandula</i> spp.	Rosmaninho ⁷	1,593	1,593	1,550	0,596
<i>Cistus salvifolius</i> L.	Sargaço ⁸	1,888	2,127	1,838	0,545
<i>Rubus</i> spp.	Silvas ⁹	0,930	0,930	0,930	0,930
<i>Dittrichia viscosa</i> (L.) W. Greuter	Tágueda ¹	1,943	2,188	-	-
<i>Ulex</i> spp.	Tojos ¹⁰	3,666	4,244	4,305	2,482
<i>Thymus vulgaris</i> L.	Tomilho ou Arçã ⁷	1,593	1,593	1,550	0,596
<i>Daphne gnidium</i> L.	Trovisco ¹	1,943	2,188	-	-
<i>Erica</i> spp. ou <i>Calluna</i> spp.	Urzes ¹¹	1,947	2,063	1,775	1,111
<i>Juniperus</i> spp.	Zimbros ¹	1,943	2,188	-	-

* Média 1 e 2 refer to mean values calculated with and without outliers, respectively.

da Silva, T.P., Pereira, J.M.C., Paúl, J.C., Santos, M.T.N., & Vasconcelos, M.J.P. (2006). Estimativa de emissões atmosféricas originadas por fogos rurais em Portugal. *Silva Lusitana*, 14(2), 239-263.

SURFACE FUEL PROPERTIES

LAYER PROPERTIES

Packing ratio (β)

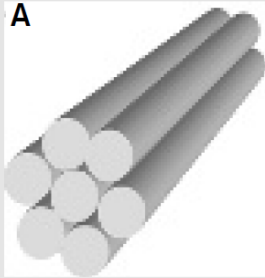
The packing ratio is an index of *fuelbed compactness*. It is given by the dimensionless ratio:

$$\beta = \frac{\rho_b}{\rho_p}, \quad \text{where} \quad \begin{array}{l} \rho_b = \text{fuel layer bulk density (kg.m}^{-3}\text{), and} \\ \rho_p = \text{particle bulk density (kg.m}^{-3}\text{)} \end{array}$$

β accounts for the *damping effect of fuel layer looseness or compression on combustion* via the fraction of fuel layer volume occupied by fuel. In a very porous, loose fuelbed (lower β) heat is lost in transmission between fuel particles, while a very compact (higher β) is poorly oxygenated. Both extremes lead to sub-optimal combustion conditions, and to reduced fire rate of spread and intensity.

SURFACE FUEL PROPERTIES

BOX 3.2. THE PACKING RATIO



The packing ratio is the proportion of fuel in a unit volume of fuelbed. The same amount of fuel can be packed (A) tightly with only 10% air or (B) loosely with 90% air.

A **high packing ratio** is unfavorable for combustion due to **poor oxygenation** of the fuel.

A **low packing ratio** is also unfavorable, because too much **heat is lost in the transmission** between widely spaced fuel particles.

Combustion efficiency is optimal at intermediate levels of the packing ratio.

CROWN FUEL PROPERTIES

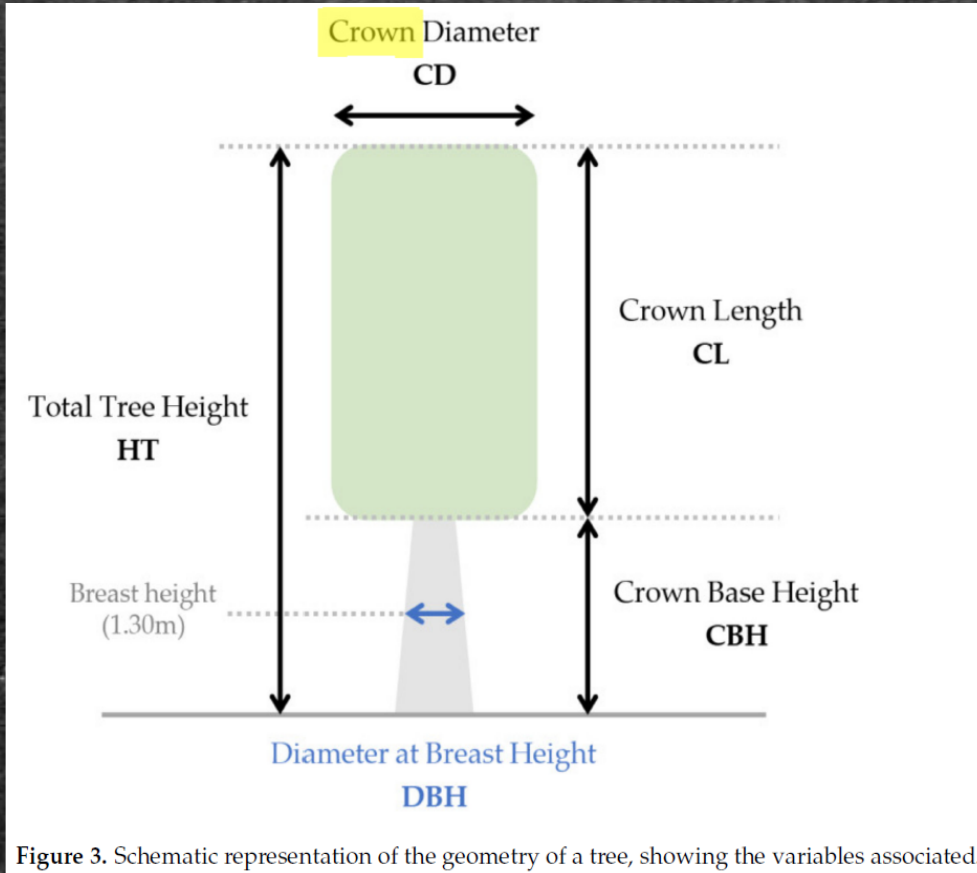


Figure 3. Schematic representation of the geometry of a tree, showing the variables associated.

CROWN FUEL PROPERTIES

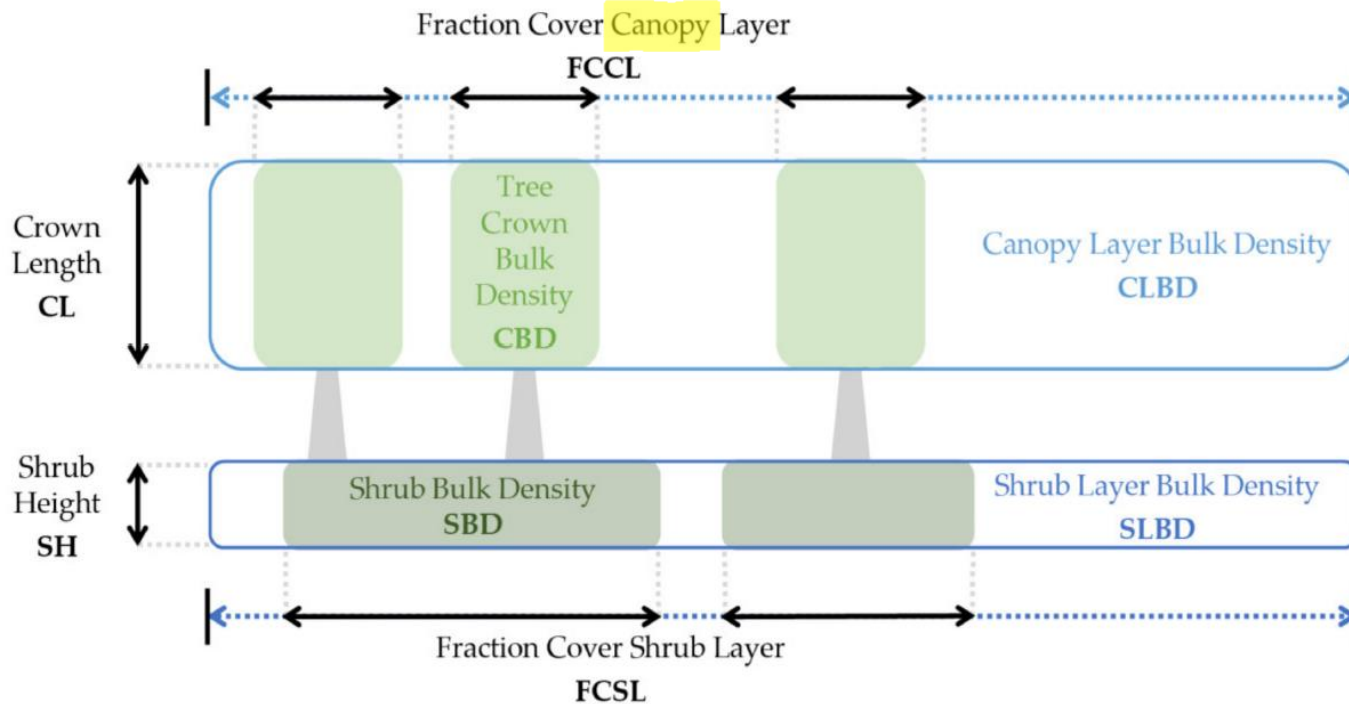


Figure 1. Schematic representation of a two-layer forest, including a tree canopy layer with height or length (CL) and a shrub layer with height (SH).

CROWN FUEL PROPERTIES

Species/Group	Sample Size		DBH (cm)	HT (m)	NHA (Trees ha ⁻¹)	G (m ² ha ⁻¹)	CBD (kg m ⁻³)	CLBD (kg m ⁻³)
Portugal	Spain							
<i>Pinus pinaster</i>	1423	7369	19.2 (10.1)	11.1 (4.2)	739 (806)	18.2 (13.3)	0.191 (0.072)	0.073 (0.060)
<i>Pinus pinea</i>	165	2152	22.2 (11.8)	8.3 (2.9)	406 (460)	11.9 (8.7)	0.195 (0.065)	0.070 (0.050)
<i>Pinus halepensis</i>	8	9264	18.5 (7.3)	7.8 (2.5)	422 (420)	9.5 (7.6)	0.073 (0.019)	0.026 (0.019)
<i>Pinus sylvestris</i>	14	6053	18.5 (7.7)	10.2 (3.6)	909 (796)	22.5 (14.2)	0.115 (0.038)	0.075 (0.039)
Other conifers	22	6312	19.4 (9.7)	10.2 (5.1)	744 (686)	18.6 (13.8)	0.048 (0.016)	0.023 (0.014)
<i>Eucalyptus globulus</i>	1059	2816	13.3 (6.8)	14.3 (4.9)	881 (964)	12.9 (11.3)	0.217 (0.101)	0.073 (0.080)
<i>Quercus suber</i>	1051	2927	24.6 (16.2)	6.5 (2.7)	244 (461)	7.1 (7.2)	0.039 (0.086)	0.011 (0.063)
<i>Quercus ilex</i> s.l.	828	10,768	25.9 (17.2)	6.3 (1.7)	356 (555)	6.7 (6.0)	0.082 (0.018)	0.027 (0.023)
<i>Quercus pyrenaica</i>	144	3201	17.6 (12.5)	8.7 (2.9)	686 (720)	11.3 (9.9)	0.010 (0.003)	0.005 (0.004)
Other oaks	58	3431	20.4 (15.9)	8.9 (4.1)	584 (669)	11.8 (10.5)	0.059 (0.021)	0.029 (0.021)
<i>Castanea sativa</i>	36	758	28.4 (25.9)	11.0 (3.4)	689 (923)	20.7 (17.4)	0.035 (0.039)	0.019 (0.020)
Other broadleaves	67	2499	27.2 (15.9)	15.1 (5.4)	584 (606)	23.1 (13.0)	0.022 (0.008)	0.017 (0.008)

Table 5. Tree species or group of species used for the estimation of the tree **Crown Bulk Density** (CBD, in kg.m⁻³) and **Canopy Layer Bulk Density** (CLBD, in kg.m⁻³), with the indication of the sample size and stand variables in the Iberian Peninsula (mean values and standard deviation in parenthesis). DBH = tree diameter at breast height; G = stand basal area.

LIDAR REMOTE SENSING OF FUELS

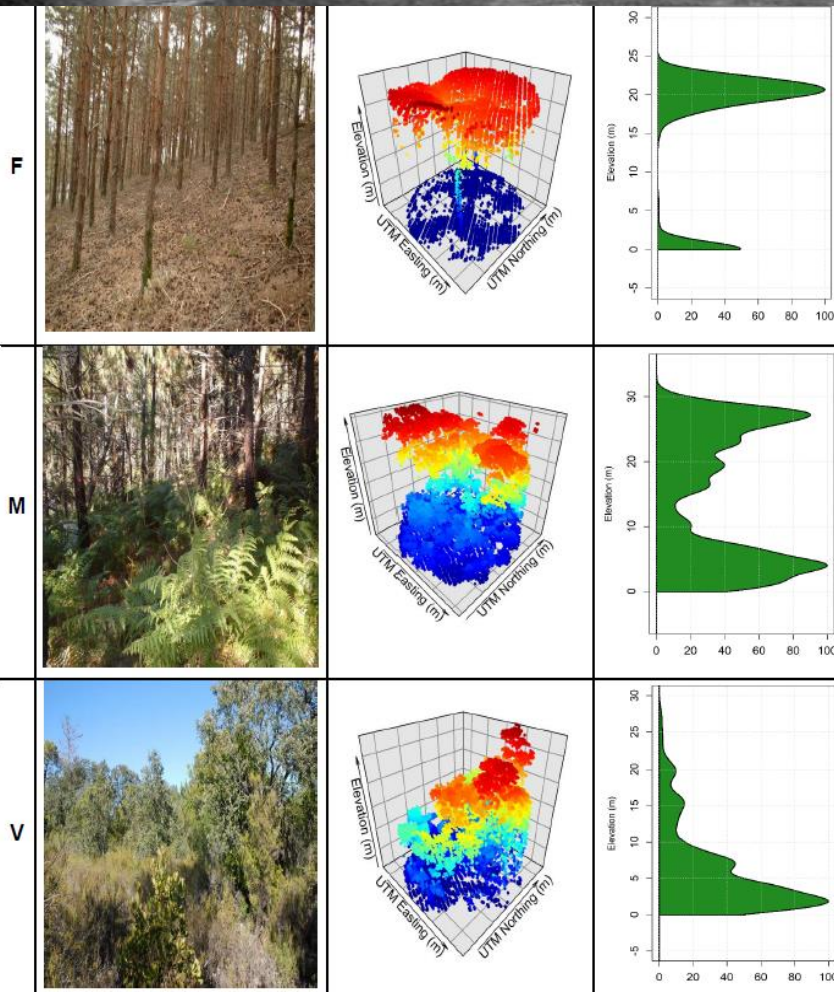
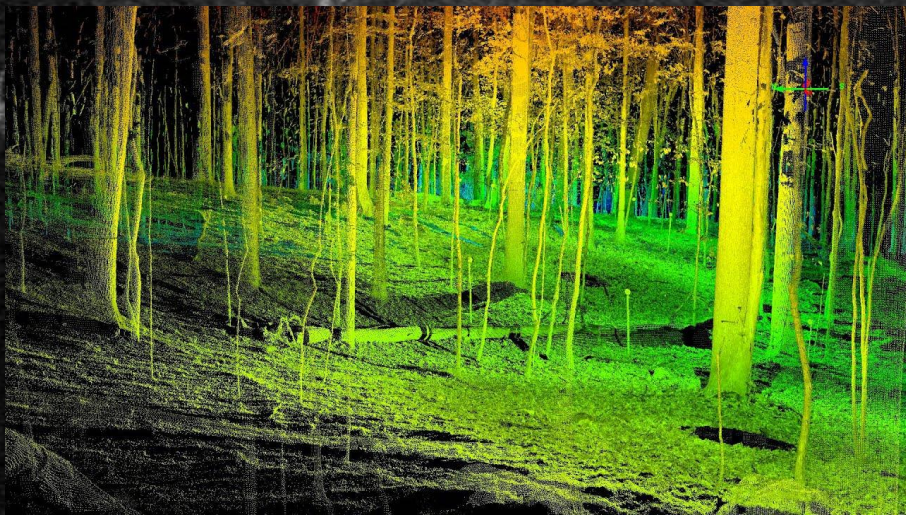
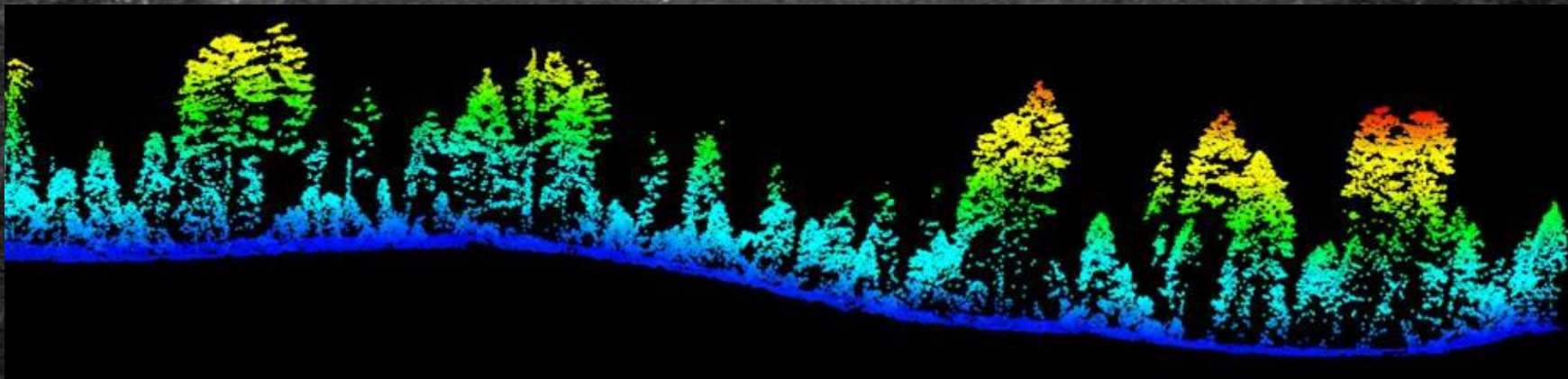
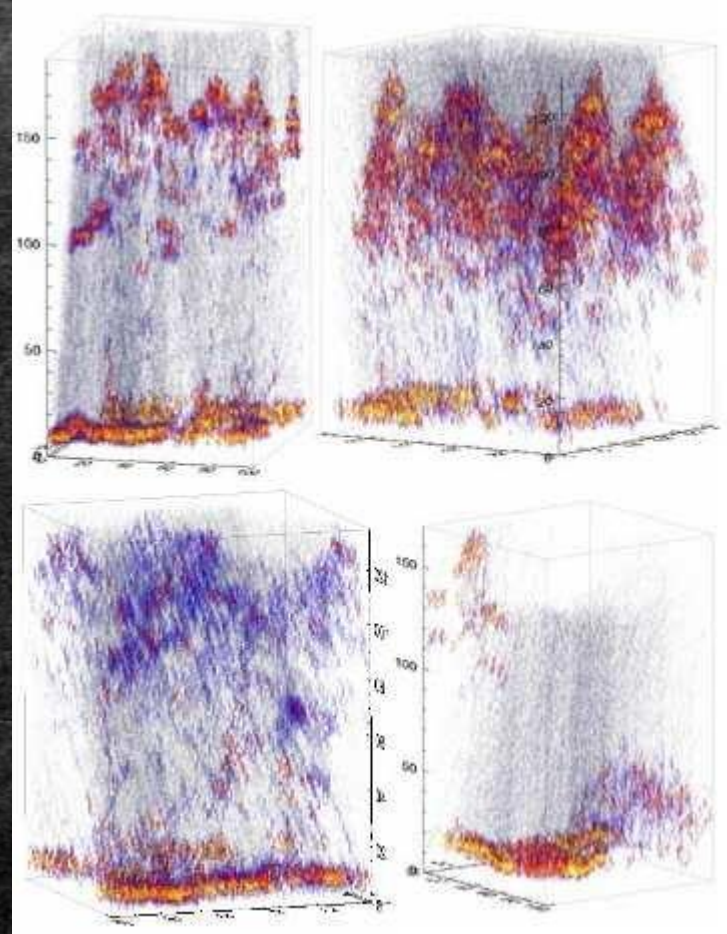
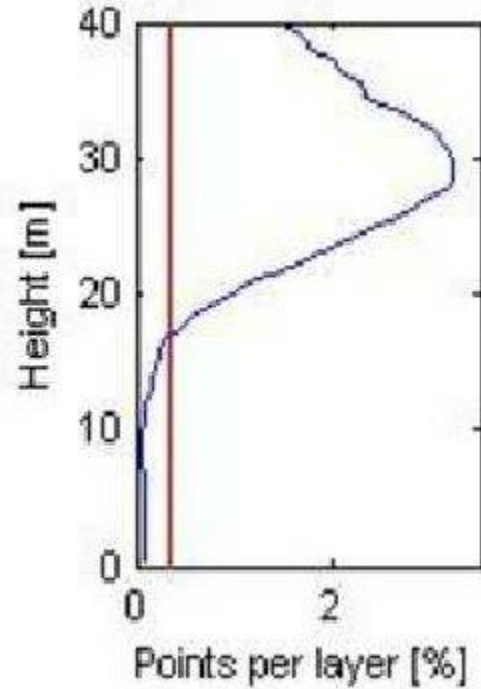
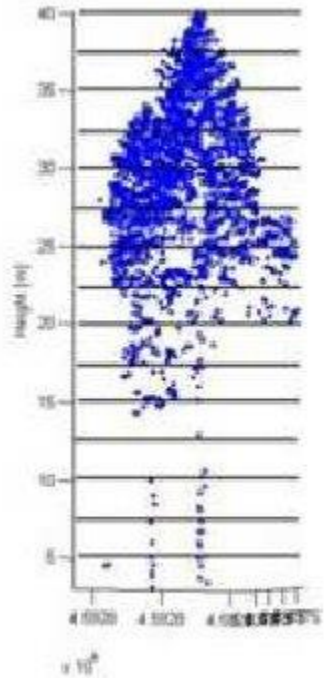


Figure 3. Three main fuel groups of fuel model in Portugal, associated with their extracted ALS point clouds and distribution of the point cloud height.

LIDAR REMOTE SENSING OF FUELS



LIDAR REMOTE SENSING OF FUELS



LIDAR REMOTE SENSING OF FUELS

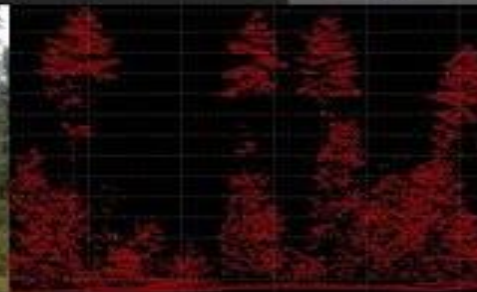
How Does LiDAR Remote Sensing Work? Light Detection and Ranging



LIDAR REMOTE SENSING OF FUELS

Measuring fire fuels with lidar

Identifying 3D Thematic Information from Lidar to Inform Fire Fuels from Tree to Region



Dr. Laura Chasmer

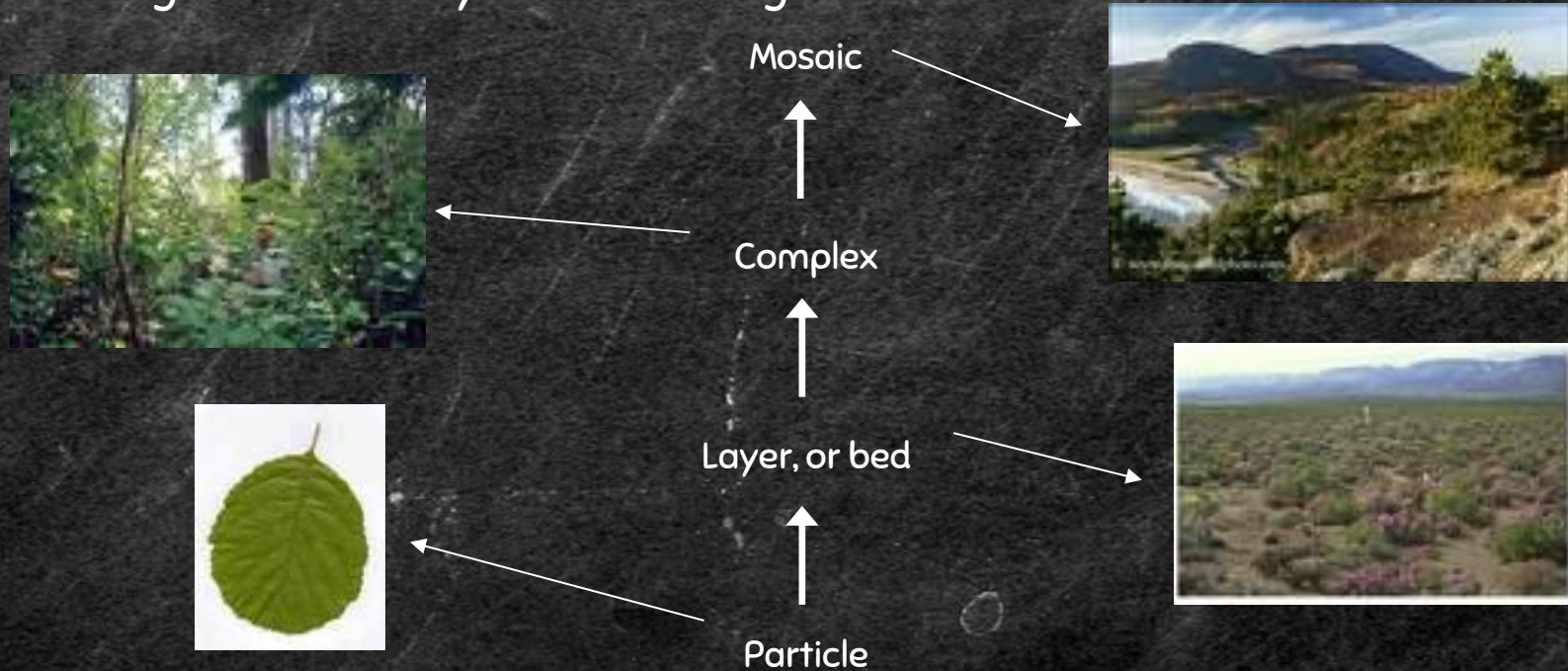
Assistant Professor
University of Lethbridge, Lethbridge Alberta



Email: laura.chasmer@uleth.ca

FUEL CLASSIFICATION AND FUEL MODELS

Fuels are described by their thermophysical and thermochemical properties according to a hierarchy of levels of organization:



FUEL CLASSIFICATION AND FUEL MODELS

The quantitative description of vegetation, considering its type, structure and status for the purpose of estimating potential fire behavior relies on ***fuel models***.

A ***fuel model*** is a simplified description of vegetation meant for use in a mathematical model of fire behavior; it is a set of numbers that describe the fuel for the model.

A set of ***18 fuel models*** were developed by Prof. P. Fernandes (UTAD) to describe the four major fuel structural groups present in Portugal:

- Litter – 4

- Mixed litter / other forest understory vegetation – 7

- Opens shrubland and shrub understory – 5

- Grasslands – 2

FUEL CLASSIFICATION AND FUEL MODELS

Fuel classification entails three tasks:

Characterization: identification of the key physical and chemical parameters to describe fuel particles, fuelbeds and fuel complexes.

Inventory: measurement or sampling of the characterization parameters.

Evaluation: estimation of ignition likelihood and potential fire behavior of the inventoried fuel. Evaluation can be strongly quantitative, relying on mathematical models and meteorological data, or qualitative, based on expert opinion.

Fuel evaluation requires specification of meteorological conditions and terrain slope.

FUEL MODELS FOR THE MAIN PORTUGUESE VEGETATION TYPES

The definition of the main structural fuel model groups is based on the *identification of the strata most responsible for the spreading of a flame front*. Table 1 shows the structural combinations of litter and vegetation, cover and height that lead to the various fuel models: litter, mixed litter+vegetation, and vegetation, under tree cover, and shrubland or grassland, in the absence of tree cover.

Table 1 – Classification matrix for fuel model groups based structure and relative importance of litter and vegetation. C = cover, h = height. D – discontinuous fuel, F – litter (*folhada*) group; M – mixed group; V – Vegetation group.

Litter	Vegetation: understory, shrubland, or pasture			
	C < 1/3	1/3 < C < 2/3	C > 2/3, h < 1 m	C > 2/3, h > 1 m
C < 3/4	D	D	V	V
C > 3/4, h < 2 cm	F	M	M	V
C > 3/4, h > 2 cm	F	M	M	M

FUEL CLASSIFICATION AND FUEL MODELS

Quantitative parameters for the Portuguese fuel models.

Model	Depth (m)	Dead fuel loading (t.ha ⁻¹)			Live fuel loading (t.ha ⁻¹)		SVR (σ, m ⁻¹)			HC (kJ.kg ⁻¹)	M _{ext} (%)
		1 hr	10 hr	100 hr	live shrubs	live herbaceous	1 hr	live herbaceous	live shrubs		
F-RAC	0.05	3.75	2	1	1.18	0	6500	-	4500	20500	28
F-FOL	0.15	2.67	1.27	0.69	1.16	0	4500	-	5000	20500	25
F-PIN	0.1	6.5	1.5	0	0	0	5500	-	-	20500	45
F-EUC	0.32	4.63	2.96	1.27	1.12	0	4200	-	5000	21000	26
M-CAD	0.63	4.54	1.87	0.61	9.08	0	6000	-	5000	20000	30
M-ESC	0.5	5.65	1.5	0.48	7.89	0	5000	-	5500	20500	25
M-PIN	0.5	7.21	3	0	6.89	0	5500	-	6000	21000	40
M-EUC	0.64	8.37	3.81	0	4.51	0	4700	-	5000	21000	32
M-EUCd	0.4	1.37	2.89	1.59	1.84	0	4500	-	5000	21000	26
M-H	0.1	2.71	1	0	0.1	0.66	5500	8000	4500	20500	30
M-F	0.3	4.5	1.5	0.5	0.48	2.35	6000	8000	4500	19500	35
V-MAb	0.5	6	0.5	0	7.5	0	4500	-	4500	21000	35
V-MAa	1.05	9.5	2.5	0	14.5	0	3500	-	4000	21000	35
V-MMb	0.9	4	0.5	0	7	0	3000	-	3000	20500	20
V-MMa	1.7	6	4	0	13	0	2500	-	3000	20500	25
V-MH	0.55	1	1	0	5.5	1.5	4500	8500	4000	19500	25
V-Hb	0.35	0.3	0	0	0	1.2	6000	6000	-	19000	24
V-Ha	0.6	0.65	0.15	0	0.4	2.35	4000	5500	4000	19000	24

characterization

inventorying



Pinus nigra
S. Padrela (V.P. Aguiar)



Pinus sylvestris
S. Alvão (V. Real)



Pseudotsuga menziesii
S. Marão (V. Real)

F-RAC

(Folhada – resinosas de agulha curta)

Densely packed **litter of short needle conifers**: *Pseudotsuga*, *Cedrus*, *Cupressus*, *Chamaecyparis*, *Pinus sylvestris*, *P. nigra* stands. Mature stands of *Acacia dealbata*.

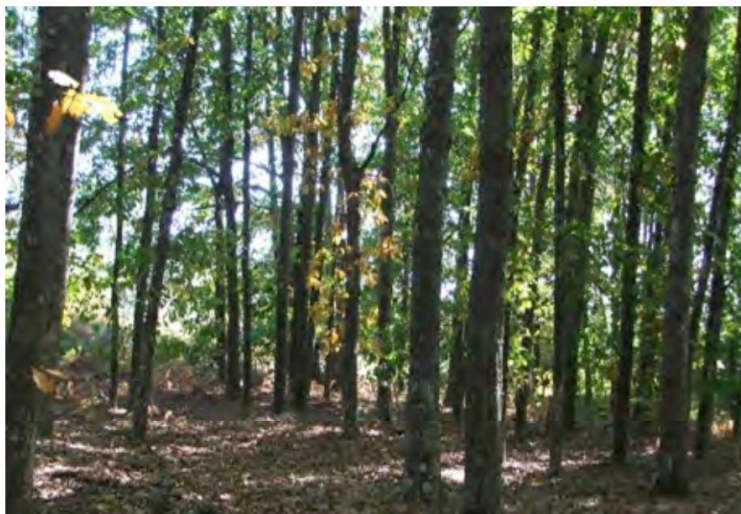
Fine fuel loading 4 – 6 t/ha.



Betula alba
S. Alvão (V. Real)



Quercus suber
Murça



Quercus rubra
S. Padrela (V.P. Aguiar)

F-FOL (Folhada – folhosas)

Densely packed litter of evergreen and deciduous hardwoods. Oak (*Quercus*), chestnut (*Castanea*), beech (*Fagus*), and birch (*Betula*) stands. Dense stands of cork oak (*Q. suber*), holm oak (*Q. rotundifolia*), strawberry tree (*Arbutus unedo*), and acacia (except *A. dealbata*).

Fine fuel loading 2 – 5 t/ha.



Eucalyptus globulus
Amarante

F-EUC

(Folhada – eucaliptos)

Eucalypt (E. globulus) leaf litter

Fine fuel loading 4 – 6 t/ha.



Pinus pinea
Alcácer do Sal



Pinus pinaster
Murça

F-PIN

(Folhada – pinheiros)

Needle litter of pine stands with intermediate length to long needles: Maritime pine (*P. pinaster*), umbrella pine (*P. pinea*), Aleppo pine (*P. halepensis*), and Monterey pine (*P. radiata*).

Fine fuel loading 4 – 7 t/ha.



Quercus pyrenaica / *Castanea sativa*
S. Marão (V. Real)



Quercus robur / *Castanea sativa*
S. Açor (Arganil)



Quercus faginea
S. Sicó (Ansião)

M-CAD

(Folhada de folhosas
caducifólias, com
mato)

**Deciduous hardwood
litter with shrub
understory** containing
abundant live fuels:
Stands of oak
(*Quercus*), chestnut
(*Castanea*), birch
(*Betula*) and beech
(*Fagus*).

Fine fuel loading 8 – 17
t/ha.



Quercus suber
Murça



Quercus suber
Mirandela



Quercus rotundifolia
Portalegre

M-ESC

(Folhada de folhosas
esclerófitas, com mato)

**Sclerophyll hardwood litter,
with shrub understory:** Cork
oak (*Q. suber*) and holm oak
(*Q. rotundifolia*) stands.

Fine fuel loading 7 – 17 t/ha.

M-PIN

(Folhada de pinheiro de agulha média a longa, com mato).

Pine litter, average to long needles and shrub understory: stands of *P. pinaster* (maritime pine), *P. pinea* (umbrella pine), *P. halepensis* (Aleppo pine), *P. radiata* (Monterey pine).

Fine fuel loading: 8–18 (t/ha)



Pinus pinaster, Sines



P. pinaster, Ponte de Lima

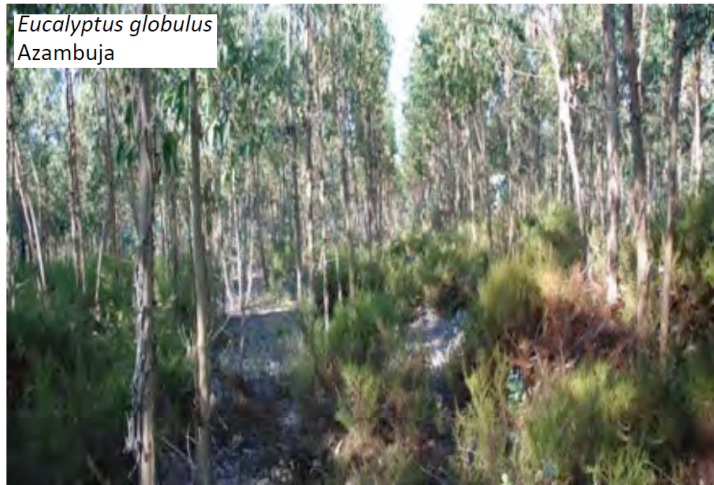


Pinus pinaster, Nisa



Pinus pinaster, Mira

Eucalyptus globulus
Azambuja



Eucalyptus globulus
Vila de Rei



M-EUC

(Folhada de eucalipto, com mato)

Eucalypt litter with shrub understory: E. globulus.

Fine fuel loading: 9–18 (t/ha)



Eucalyptus globulus
Sever do Vouga



Eucalyptus globulus
Celorico de Basto

M-EUCd

(Folhada descontínua de eucalipto, com ou sem mato)

***Eucalypt patchy litter*, with or w/out shrub understory along plantation rows: young or recently *E. globulus*.**

Fine fuel loading: 1–4 (t/ha)



Betula alba, Sta. Marta Penaguião



Eucalyptus globulus,
V. Velha Rodão



Pinus pinea, Alcácer do Sal

M-H

(Folhada com erva, qualquer espécie arbórea)

Litter, with grass understory:
forest stands, regardless of species.

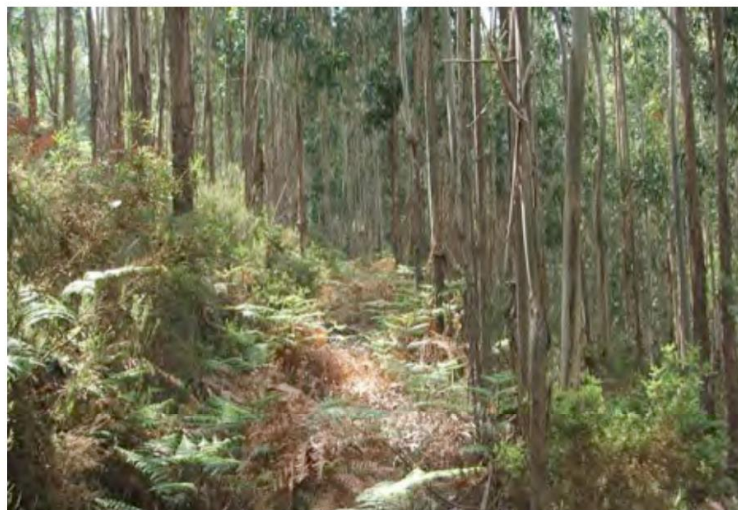
Fine fuel loading: 2–5 (t/ha)



Quercus pyrenaica, Vila Real



Pinus pinaster, Vila Real



Eucalyptus globulus,
Celorico de Basto

M-F

(Folhada com fetos, qualquer espécie arbórea)

**Litter, with bracken fern
understory:** forest stands,
regardless of tree species.

Fine fuel loading: 6–9 (t/ha)



Erica spp. / *Pterospartium*, Amarante



Erica spp. / *Pterospartium*, V. Real



Erica australis / *Pterospartium*
Vila Real

V-MAb (Mato baixo)

Low shrubs (<1 m) with abundant dead and/or fine fuel. Heath (*Erica*), gorse (*Ulex*), prickly broom (*Chamaespartium tridentatum*), and juniper (*Juniperus*) shrublands. Also as forest understory in open or young stands, regardless of tree species.

Fine fuel loading: 7–14 (t/ha)



Ulex europaeus, Ilha da Madeira



Erica scoparia, Mafra



Ulex europaeus / *P. pinaster*
V. Real

V-MAa

(Mato alto)

Tall shrubs (>1 m) with abundant dead and/or fine fuel. Heath, gorse, prickly broom, and other old broom shrublands.

Also as forest understory in open or young stands, regardless of tree species. Dense natural maritime pine regeneration.

Fine fuel loading: 12–27 (t/ha)



Cytisus, Manteigas



Quercus coccifera, Porto de Mós



Cistus salvifolius, Almodovar

V-MMb

(Mato baixo, pouca biomassa morta)

Low shrubs (<1 m), with little dead fuel and/or relatively thick foliage.

Genista, *Cytisus*, rock rose, kermes oak, wild olive, strawberry tree, lentisk, and other mediterranean species shrublands. Blackberry thickets. Also as forest understory in open or young stands, regardless of tree species.

Fine fuel loading: 4–8 (t/ha).

V-MMa

(Mato alto, pouca biomassa morta)

Tall shrubs (>1 m), with little dead fuel and/or relatively thick foliage.

Genista, Cytisus, rock rose, kermes oak, wild olive, strawberry tree, lentisk, and other mediterranean species shrublands. Blackberry thickets. Also as forest understory in open or young stands, regardless of tree species.

Fine fuel loading: 10–19 (t/ha)



Cytisus multiflorus /*Lavandula*, Celorico da Beira



Q. rotundifolia/ *Arbutus unedo*/ *Er. scoparia*/ *Cistus*/*Lavandula*, Fátima

Cistus ladanifer, Mirandela



Quercus coccifera, Porto de Mós



Pterospartium, Mesão Frio



V-MH

(Mato baixo e verde, com erva)

Short (<1 m), green, often patchy shrubs, with grasses.

Young shrublands, up to three years since the last fire.

Fine fuel loading: 1–5 (t/ha).

Pterospartium / *Halimium* / *Erica*
Castro Daire



P. pinea / *P. pinaster*, Condeixa-a-Nova



V-Hb

(Erva baixa)

Short grass (<0,5 m).

Meadows, pastures, recent fallow lands, cereal stubble, evergreen oak woodlands.

Fine fuel loading: 1–1.5 (t/ha).



Quercus pyrenaica, Marvão.



V-Ha

(Erva alta)

Tall grass (> 0,5 m).

Meadows, pastures, recent fallow, cereal fields, reed, evergreen oak woodlands.

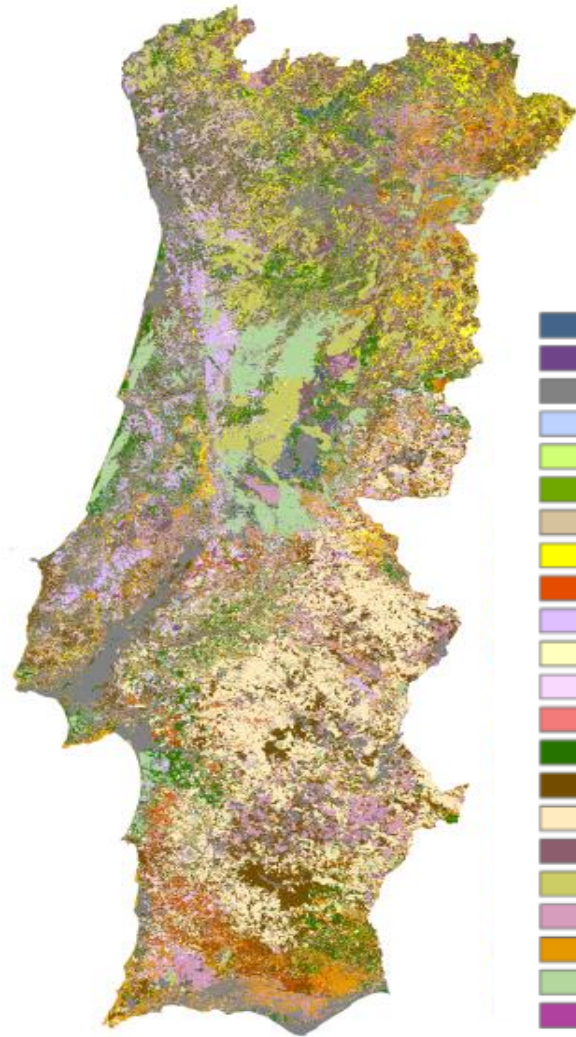
Fine fuel loading: 2–4 (t/ha).

FUEL CLASSIFICATION AND FUEL MODELS

Fuel model distribution (%) by forest type in Portugal (NFI4, 2005–06)
Dominant fuel models in each forest type in **bold**.

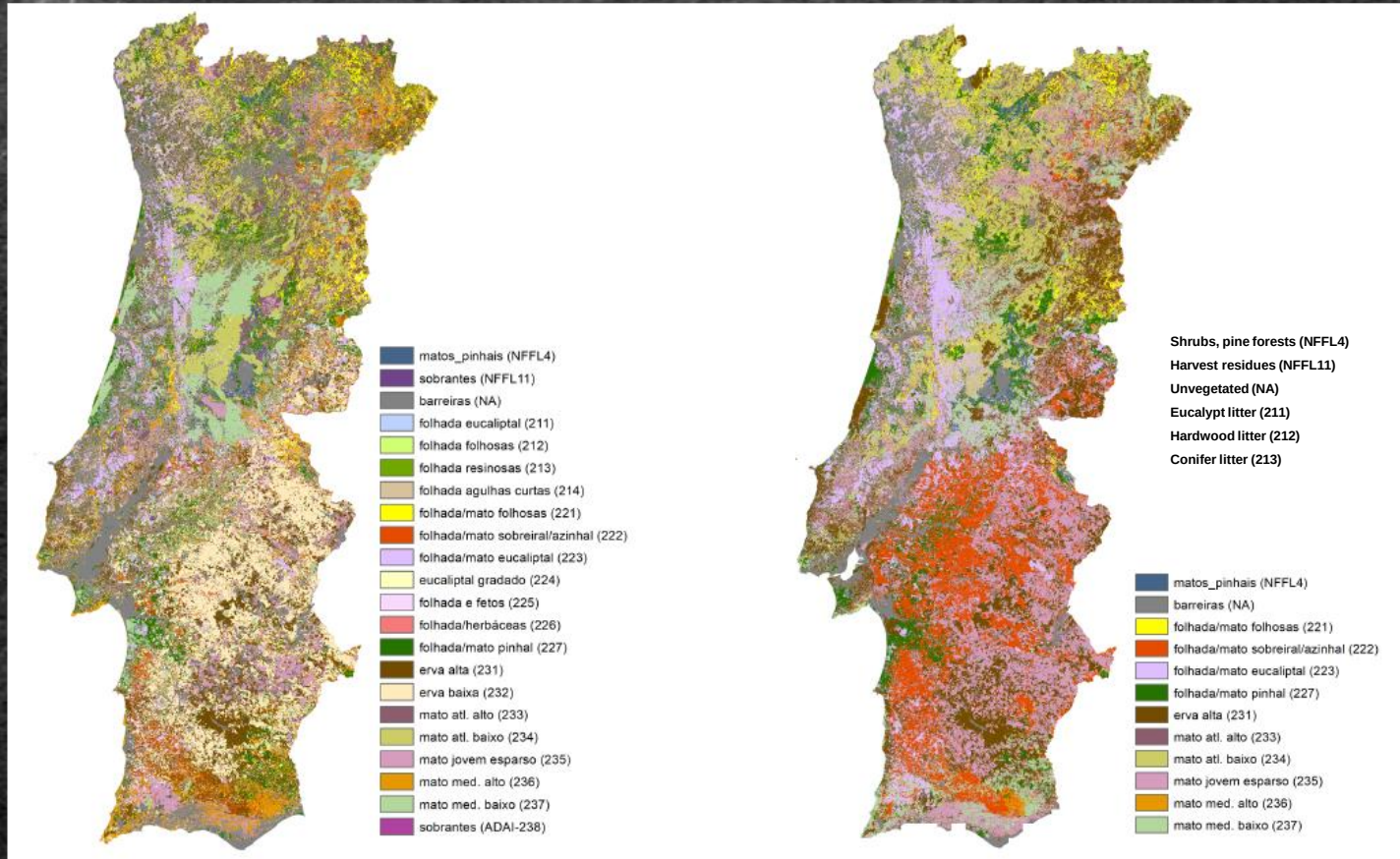
Model	Holm oak	Other oak	Eucalypt	Other hardwood	Other softwood	Maritime pine	Umbrella pine	Cork oak
F-RAC	0	0	0	0	17.9	0	0	0
F-FOL	1.2	10.4	0	8.3	0	0	0	7.6
F-PIN	0	0	0	0	3.6	14	9.5	0
F-EUC	0	0	6	1.9	0	0	0	0
M-CAD	0	31.1	0	13.9	0	0	0	0
M-ESC	2	0	0	0	0	0	0	9
M-PIN	0	0	0	0	7.1	41	11.9	0
M-EUC	0	0	22.9	14.8	0	0	0	0
M-EUCd	0	0	36.5	0	0	0	0	0
M-H	0.7	1.4	0.9	0.9	0	0.6	2.4	1.4
M-F	0.2	0.9	0.7	2.8	3.6	1.5	0	0.2
V-MAb	0.2	0.5	2.6	2.8	3.6	4.1	9.5	9.4
V-MAa	0	0	3	0.9	3.6	2.7	2.4	2
V-MMb	40.4	18.4	9.4	22.2	28.6	14.6	26.2	38.3
V-MMa	28.7	33.5	17.3	25	32.1	19.8	31	20.6
V-Hb	18.6	3.3	0.3	5.6	0	0.9	4.8	9
V-Ha	8.1	0.5	0.4	0.9	0	0.8	2.4	2.6

FUEL MODEL MAP FOR PORTUGAL (2018)



- Pine forest shrubs (NFFL4)
- Forest harvest residues (NFFL11)
- Unvegetated (NA)
- Eucalypt litter (211)
- Hardwood litter (212)
- Long needle conifer litter (213)
- Short needle conifer litter (214)
- Hardwood litter/shrubs (221)
- Evergreen oak litter/shrubs (222)
- Eucalypt litter/shrubs (223)
- Harrowed eucalypt stands (224)
- Litter/bracken fern (225)
- Litter/grasses (226)
- Pine litter/shrubs (227)
- Tall grass (231)
- Short grass (232)
- Tall atlantic shrubs (233)
- Short atlantic shrubs (222)
- Sparse young shrubs (235)
- Tall mediterranean shrubs (236)
- Short mediterranean shrubs (236)
- Harvest residues (ADAI238)

FUEL MODEL MAPS FOR PORTUGAL



FIRE, DECOMPOSITION AND LEAF TRAITS

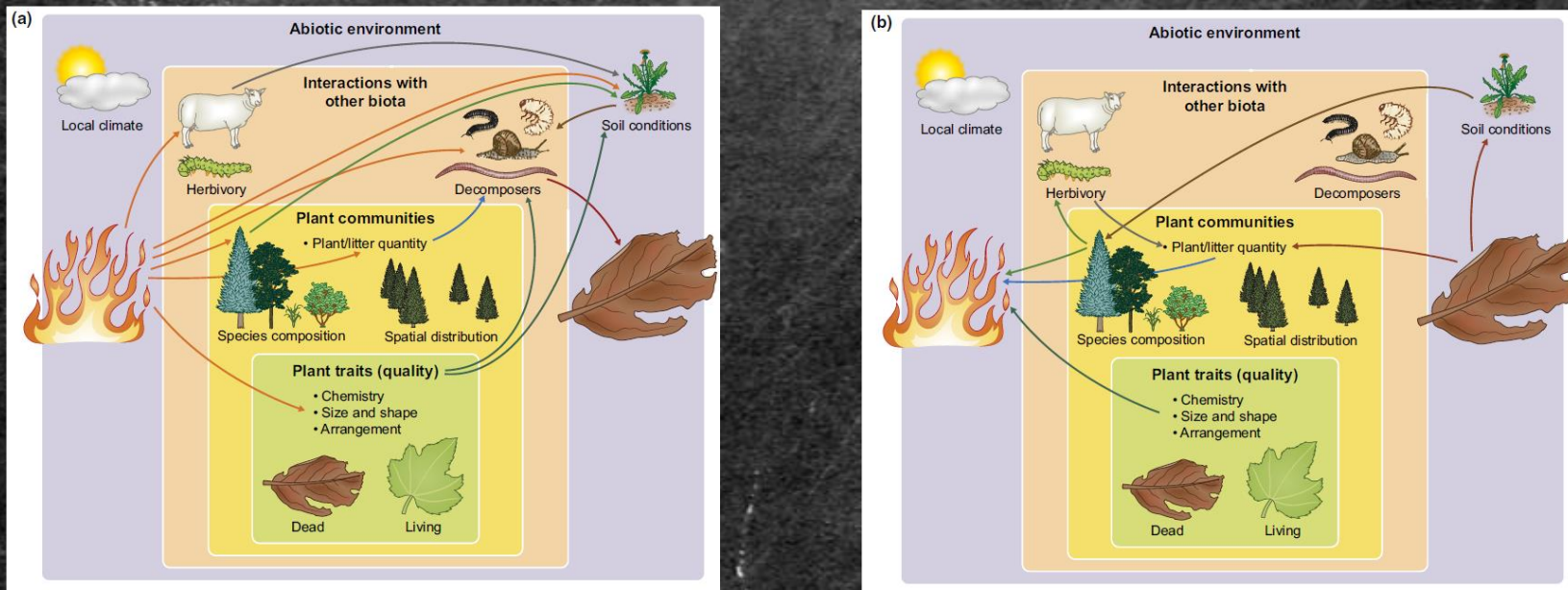


Fig. 1 Fire–decomposition feedbacks with a central role for plant traits of different species, adding concentric shells of real-world complexity through community-level interactions and other biotic and abiotic drivers. For clarity the two directions of these feedbacks are separated here: **(a) fire effects on decomposition** and **(b) decomposition effects on fire**. The **core** represents the **role of plant traits**; the **shell** around it represents **ecosystem-level factors** such as **community composition** and **species (litter) interactions**, and **amount and spatial configuration of vegetation and litter** (fuel). The next shell adds **other biotic interactions** while the outer shell represents the **(local to global) abiotic drivers** that may affect processes and connections within the inner shells.